

Developing banking credit scorecards

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1 Documentation and Audit trail

During the statistical study, the developer has to:

- Save all processing files in a dedicated directory, including original and intermediate datasets, programs (with detailed comments), Python functions or SAS macros, Python/R version and related packages.
- If strongly advised that the auditor be able to run the scripts associated with this scoring methodology in their own environment for a better understanding and validation of the results, saving and versioning using a Git repository (Git, GitHub, GitLab, etc.) is recommended.
- Continue a study report, which contains:
 - Definition of the purpose;
 - Scope of the model;
 - Application perimeter;
 - References to all saved processing files;
 - Key outputs and related comments;
 - Explanation of all processing choices;
 - Link to all external working documents;
 - Explanation of any planned point to be carried out;
 - Comments on model limitations and identified risk factors.

The model files and the study report constitute the audit trail.

At the end of the study, the statistician must create a durable archive in a dedicated directory.

The archive should contain:

- the study report;
- all documents referenced in the report;
- all processing files referenced in the study report (dataset, programs, outputs).

2 Study Dataset Definition and Preparation

This chapter describes the recommended process for constructing the study dataset used for credit scorecard development. The objective is to ensure that the dataset is reproducible, auditable, statistically representative and operationally deployable.

2.1 Objectives and Governance

The study dataset should be built independently from the modelling phase and fully documented. At a minimum, documentation should describe the observation unit, observation period, source systems, extraction criteria, assumptions, transformations and version information.

All extraction scripts, intermediate datasets, parameters and validation outputs should be archived to guarantee full reproducibility.

2.2 Data Extraction and Preparation

The initial dataset should include every variable that may potentially be useful for modelling.

When several information systems are involved, the methodology should document:

- source systems and business owners;
- extraction dates and selection criteria;
- loading procedures;
- join keys and join types;
- reconciliation and consistency checks.

Typical quality controls include row counts, duplicate detection, missing keys, join coverage and comparison of aggregated figures before and after data integration.

Any extraction issue (missing data, corrupted files, inconsistent identifiers, late updates or incomplete history) should be documented together with its expected impact on model development.

2.3 Study Population

The modelling population must reflect the future population on which the score will be applied.

The study should therefore specify:

- business objectives of the score;
- products, markets and channels covered;
- excluded populations;
- operational deployment constraints.

Whenever multiple applications from the same customer exist, the treatment adopted should be documented to justify the independence assumption required by many statistical models.

2.4 Variable Assessment

Each candidate predictor should be evaluated from four complementary perspectives:

1. business relevance;
2. statistical quality;
3. operational availability;
4. regulatory compliance.

The variable inventory should contain, whenever applicable:

- business definition and source;
- missing-value rate;
- distribution and outlier diagnostics;
- stability over time;
- availability at application date;
- inclusion or exclusion rationale.

Variables unavailable at scoring time, unstable over time or prohibited by regulation should normally be excluded.

Missing values should be analysed individually and handled using an appropriate strategy (dedicated category, statistical imputation, observation removal or variable exclusion), with the expected impact on model performance documented.

2.5 Target Definition

The target variable should represent the business event that the score is expected to predict.

For application scorecards, a binary target is generally defined as

$$QUAL = \begin{cases} 1 & \text{Bad customer} \\ 0 & \text{Good customer} \end{cases}$$

The target should rely on a single, stable and interpretable risk criterion, aligned with business objectives. Its construction must specify:

- default definition;
- observation horizon;
- excluded contracts;
- calculation rules.

If the target cannot be computed reliably, the observation should be excluded or flagged as missing.

2.6 Observation Window

The observation window should provide sufficient historical depth while remaining representative of current business practices.

Its definition should balance:

- sample size;
- target maturity;
- portfolio representativeness;
- stability of underwriting policies;
- seasonality effects.

For revolving products, activation behaviour should be analysed separately when application and activation dates differ.

2.7 Development and Validation Samples

Once the modelling population has been finalised, the dataset should be split into independent development and validation samples.

The splitting procedure should be reproducible, statistically representative and fully documented.

2.8 Dataset Quality Summary

Before model development begins, a final summary should confirm:

- population size;
- bad rate;
- representativeness of the modelling sample;
- distributions of key variables;
- data-quality indicators;
- major assumptions and limitations.

These analyses constitute the baseline documentation supporting subsequent feature engineering, model estimation, validation and regulatory review.

3 Study dataset definition and preparation

3.1 Purpose and auditability principles

This chapter defines the minimum requirements to build the study dataset used for scorecard development.

The objective is to ensure that dataset preparation is:

- reproducible;
- auditable;
- independent of the study itself;
- clearly documented.

Version control and reproducibility are strongly recommended.

Audit evidence must be retained in a dedicated archive including:

- original extracts;
- intermediate datasets;
- scripts and programs;
- execution parameters;
- run logs;
- output figures and tables.

3.2 Data Extraction and Initial Dataset Construction

3.2.1 Initial dataset definition

The first step consists in building the initial dataset.

This dataset should contain all variables potentially useful for the modelling exercise.

The initial dataset must be described precisely:

- observation unit;
- observation period;
- transformations;
- data quality assumptions;
- identifiers;
- target availability.

3.2.2 Source systems and extraction

When several source systems are involved, the following information must be documented:

- source system(s);
- table names or extraction views;
- extraction date;
- extraction criteria;
- ownership and business responsibility.

The extraction must be reproducible.

Evidence that extracted values correctly reflect production data should be available.

3.2.3 Transformations and joins

If multiple datasets are combined, the methodology must specify:

- loading procedures;
- join keys;
- join types;
- expected coverage;
- handling of non-matches;
- reconciliation checks.

Typical controls include:

- row counts;
- duplicates;
- missing keys;
- key distributions;
- consistency of aggregated figures.

3.2.4 Data issues and impact analysis

Any encountered issue must be documented:

- missing values;
- corrupted files;
- inconsistent identifiers;
- incomplete history;
- late updates;
- extraction failures.

The impact of each issue should be assessed on:

- model building;
- risk indicators;
- representativeness;
- target computation.

3.3 Study Scope and Intended Use of the Score

3.3.1 Score purpose and business objectives

The score must be defined using the following minimum dimensions:

- Stakeholders and ownership;
- Business objectives;
- Expected use of the score;
- Model renewal rationale, when applicable.

Examples of business objectives are:

- default prediction;
- risk reduction;
- limit assignment;
- portfolio management;
- collection prioritisation.

3.3.2 Deployment constraints and application perimeter

Deployment considerations should be addressed upfront.

The following items must be documented:

- target products;
- markets and channels;
- decision process integration;

- expected deployment timeline;
- system constraints;
- operational limitations;
- populations excluded from the score.

Audit evidence to retain:

- deployment objectives;
- environment description;
- limitations;
- assumptions retained in the study.

3.4 Observation Perimeter and Exclusion Criteria

3.4.1 Perimeter exclusions

Some applications may need to be excluded from the study perimeter.

Examples:

- applications for which the score will not be applied;
- defaults not detected by the score;
- applications that will disappear in the future;
- operational exceptions.

Depending on the case, exclusions may apply:

- only to modelling;
- only to cut-off computation;
- to both modelling and cut-off construction.

3.4.2 Statistical independence assumption

When logistic regression is used, the independence assumption must be assessed.

If multiple applications per customer exist, the treatment chosen must be documented and justified.

Audit evidence:

- observation exclusion rules;
- impact assessment;
- associated programs;
- statistics before and after filtering.

3.5 Initial Variable Analysis and Selection

3.5.1 Variable inventory and selection table

A variable inventory table must be built.

For each variable:

- source and lineage;
- business definition;
- missing rate;
- mode and concentration;
- distribution diagnostics;
- outlier detection;
- inclusion or exclusion rationale.

Variables deemed not informative or unsuitable for operational deployment may be excluded from the modelling exercise.

3.6 Data Coherence, Quality and Compliance Assessment

3.6.1 Coherence checks

When possible, coherence between sources should be checked.

Examples include:

- comparison of the same variable obtained from different systems;
- consistency between dates and derived variables;
- comparison between aggregated indicators;
- cross-checks using external references.

Whenever inconsistencies are detected:

- the issue must be documented;
- the impact on modelling must be assessed;
- a remediation plan must be proposed.

3.6.2 Operational and control constraints

The following aspects should be analysed for each candidate variable:

- retrieval cost and operational feasibility;
- reliability and robustness;
- business meaning and stability over time;
- availability before the decision process;
- operational deployment constraints.

Variables unavailable at scoring time should normally be excluded.

3.6.3 Regulatory and privacy constraints

Variables must comply with local regulatory requirements.

Examples include:

- GDPR in Europe;
- local banking secrecy rules;
- data minimisation principles;
- restrictions on sensitive variables.

The Data Privacy Expert should be consulted whenever necessary.

Documentation must explicitly mention:

- variables excluded for regulatory reasons;
- mitigation measures;
- remaining limitations.

3.6.4 Missing values handling

For each variable containing missing values, the chosen treatment must be documented and justified.

Possible treatments include:

- missing as a separate category;
- mean or median imputation;
- mode imputation;
- exclusion of the variable;
- exclusion of observations.

The impact on model stability and performance should be assessed.

3.7 Risk Indicator Definition and Computation

If historical information exists, the risk indicator must be computed using:

- defaults;
- incidents;
- restructurings;
- catch-up plans;
- rescheduling events;
- other relevant indicators.

The methodology should specify:

- which contracts are excluded;
- which events define a “seriously bad” customer;
- observation windows;

- calculation rules.

Audit evidence must include:

- computation programs;
- indicator definitions;
- consistency checks;
- validation outputs.

3.8 Target Definition (Risk Criterion)

3.8.1 Target definition principles

The objective is to define a binary target variable classifying loans into:

- Good;
- Bad.

The target should be:

- simple;
- stable over time;
- aligned with business objectives;
- observable over a sufficiently long period.

A recommended approach is:

1. Define a “seriously bad” outcome over an observation period L ;
2. Select a criterion computed over a shorter period H that best predicts this outcome.

3.8.2 Single-criterion requirement

The target should be based on a single risk criterion.

Mixing different risk criteria is discouraged because it may:

- reduce interpretability;
- increase instability;
- weaken business understanding.

3.8.3 Convention

The target variable is denoted:

$$\text{QUAL} = \begin{cases} 1 & \text{Bad customer} \\ 0 & \text{Good customer} \end{cases}$$

If the target cannot be computed, the observation should be flagged as missing.

Audit evidence:

- analysis notebooks;
- target construction programs;
- stability analyses;
- documented rationale.

3.9 Definition of the Observation and Modelling Time Window

The time-window must be designed so that:

- the target can be computed for almost all financed loans;
- the observation horizon is representative of future behaviour;
- seasonality effects are limited;
- sample size remains sufficient.

The beginning date should balance:

- recency;
- representativeness;
- stability of business rules.

3.9.1 Validation of target and time-window definitions

The target definition and time-window must be validated through:

- bad rate stability;

- representativeness analyses;
- sensitivity analyses;
- business reviews.

If the bad behaviour is insufficiently represented:

- extend the observation window;
- revise the target definition;
- document the limitations.

3.9.2 Revolving credit specificities

For revolving credit products:

- activation date may differ from application date;
- activation behaviour must be analysed;
- the time between application and activation should be monitored;
- activation rates may impact representativeness.

Audit evidence to retain:

- activation analyses;
- associated programs;
- business interpretation.

3.10 Loan Class Definition and Population Structuring

3.10.1 Loan population classification

The portfolio should be split into several categories:

- Accepted and financed loans;
- Loans with unknown QUAL;
- Rejected applications;
- Applications without follow-up;

- Operational exceptions.

This classification is essential for:

- modelling;
- cut-off construction;
- reject inference;
- portfolio monitoring.

3.10.2 Rejection rules evolution and simulation

Rejection rules may evolve over time.

When historical rules differ from current rules:

- future rules should be simulated;
- impacts on portfolio composition should be assessed;
- resulting biases should be documented.

3.11 Development and Validation Dataset Construction

Two distinct datasets are generally created:

- Development dataset: used to estimate model parameters;
- Validation dataset: used exclusively for performance assessment.

The split should be:

- reproducible;
- documented;
- statistically representative.

3.12 Dataset Summary and Representativeness Overview

A summary table should present:

- population sizes;

- bad rates;
- distributions of key variables;
- proportions of financed and rejected loans;
- representativeness metrics.

3.13 Baseline Metrics and Acceptance Monitoring

Acceptance monitoring should be built on the study dataset in order to:

- improve portfolio understanding;
- check consistency with historical behaviour;
- compare current production with previous scorecard generations;
- detect unexpected drifts.

Suggested KPIs include:

- weighted average score of financed loans;
- acceptance rate;
- override rate;
- bad rate among financed customers;
- risk rate among rejected applications;
- portfolio composition.

Audit evidence to retain:

- monitoring programs;
- KPI definitions;
- output tables and figures;
- interpretation included in the study report.

4 Reject Inference

4.1 Reject inference through expert policy engineering

This section describes a recommended reject inference framework for cut-off setting and portfolio simulation.

Unlike traditional reject inference techniques, this methodology does not try to estimate the unknown good/bad outcome for previously rejected applications.

Instead, pseudo-observations are generated under explicit expert assumptions.

The methodology relies on two components:

- the statistical score, whose role is to rank applications according to relative risk;
- expert assumptions describing the proportion of rejected applications that would be financed under a new policy and their relative risk.

Under this framework:

- reject inference is independent from score development;
- the method is transparent and auditable;
- assumptions are explicitly documented;
- the framework is suitable for business discussions and governance.

4.1.1 Rationale

A classical issue in reject inference is that the financed population is not representative of the entire application flow.

If this selection bias is ignored:

- estimated bad rates may be biased;
- cut-offs may be sub-optimal;
- portfolio simulations may become unrealistic.

Traditional reject inference methods attempt to estimate the missing targets for rejected applications.

The present framework avoids this issue by:

- explicitly modelling business assumptions;
- separating statistical modelling and policy assumptions;
- documenting all expert choices.

This approach offers several advantages:

- easier communication with stakeholders;
- stronger governance;
- improved auditability;
- more flexible scenario analysis.

4.1.2 Scope and population split

The methodology applies to the full application flow.

Applications are split into several operational populations:

- A: financed applications with observed performance;
- B: financed applications for which the target cannot yet be computed;
- C: applications rejected because of risk;
- D: applications rejected for operational reasons;
- E: applications rejected for other reasons.

These operational definitions must be consistent with:

- local credit processes;
- portfolio governance;
- the cut-off definition.

4.1.3 General principle

The score is segmented into score brackets.

For each bracket:

- the observed bad rate is computed;

- expert assumptions are applied to rejected populations;
- a simulated portfolio is obtained;
- cumulative risk and acceptance rates are calculated.

Two types of assumptions are introduced:

1. Selection assumptions:

What proportion of each rejected population would have been financed under the new policy?

2. Risk assumptions:

How risky are these populations compared to financed applications?

The reference bad rate is computed from the financed population.

For each non-observed population k , the assumed risk is:

$$r_k = w_k \times r^{obs}$$

where

- r^{obs} is the observed bad rate;
- w_k is an expert multiplier.

4.1.4 Calculation method

For a given score bucket b , define:

- A_b : number of financed applications with observed performance;
- Bad_b : number of bad financed applications;
- C_b, D_b, E_b : sizes of rejected populations;
- w_C, w_D, w_E : risk multipliers.

The adjusted financed volume is:

$$N_b^{adj} = A_b + w_C C_b + w_D D_b + w_E E_b$$

The adjusted number of bads is:

$$Bad_b^{adj} = Bad_b + w_{CrC}C_b + w_{DrD}D_b + w_{ErE}E_b$$

The adjusted bad rate is:

$$R_b^{adj} = \frac{Bad_b^{adj}}{N_b^{adj}}$$

The cumulative indicators above a cut-off c are:

$$Risk(c) = \frac{\sum_{b \geq c} Bad_b^{adj}}{\sum_{b \geq c} N_b^{adj}}$$

and

$$Acc(c) = \frac{\sum_{b \geq c} N_b^{adj}}{\sum_b N_b^{adj}}$$

These indicators are used to calibrate the operational cut-off.

4.1.5 Operational completion of the expert table

The expert table documents:

- selection weights;
- risk multipliers;
- business rationale;
- benchmark information;
- supporting evidence.

The table should remain:

- transparent;
- auditable;
- reproducible;
- easy to update.

4.1.6 How to complete the expert table

The following steps are recommended:

1. Confirm the population perimeter;
2. Compute the observed reference bad rate;
3. Select financing proportions for rejected populations;
4. Define risk multipliers;
5. Derive implied risks;
6. Document assumptions;
7. Review cut-off impacts;
8. Validate and archive assumptions.

All assumptions should be supported by:

- business expertise;
- historical observations;
- benchmark studies;
- prudential considerations.

4.1.7 Interpretation of the expert parameters

The expert coefficients do not correspond to model parameters.

They correspond to business assumptions and should be interpreted as follows:

- w_k expresses the expected proportion of population k that would effectively be financed under the simulated policy;
- u_k expresses how much more or less risky population k is expected to be relative to the observed financed population.

Examples:

- $w_D = 0.10$ means that only 10% of previously rejected applications belonging to population D would be financed under the new policy;

- $u_D = 0.80$ means that the bad rate of these applications is assumed to be 20% lower than the observed financed population;
- $u_D = 1.20$ means that the population is assumed to be 20% riskier than the financed reference population.

The expert assumptions are not intended to predict individual outcomes but to provide realistic portfolio simulations.

4.1.8 Monitoring of assumptions over time

Because expert assumptions rely on business judgement, they must be monitored over time.

Monitoring objectives include:

- verifying consistency with observed portfolio behaviour;
- ensuring assumptions remain aligned with business strategy;
- detecting obsolete assumptions;
- challenging overly conservative or optimistic scenarios.

The following indicators may be monitored:

- stability of observed reference risk;
- stability of score-bucket structures;
- financing proportions by segment;
- sensitivity of cut-off decisions to assumptions;
- comparison with benchmark studies.

When deviations are detected:

- assumptions should be reviewed;
- supporting evidence should be updated;
- the expert table may be recalibrated.

4.1.9 Expert justification requirements

Each expert assumption must be formally documented.

At a minimum, the documentation should specify:

- the population concerned;
- the coefficient value;
- the business rationale;
- the supporting evidence;
- the source of information;
- the date of validation;
- the validating expert or committee.

Possible sources of evidence include:

- historical observations;
- benchmark studies;
- business analyses;
- pilot campaigns;
- prudential assumptions.

The objective is to ensure that:

- assumptions are transparent;
- assumptions are challengeable;
- assumptions are auditable;
- assumptions can be revised when necessary.

4.1.10 Relationship with the scoring-based reject inference approach

The expert-based reject inference approach is complementary to statistical reject inference techniques.

However, it should not be considered the default approach for score construction.

Its main strengths are:

- interpretability;
- transparency;
- governance;
- ease of implementation;
- sensitivity analysis capabilities.

By contrast, scoring-based reject inference approaches:

- attempt to infer the missing targets;
- rely on stronger statistical assumptions;
- may provide more granular estimates;
- require additional validation efforts.

4.1.11 Archiving and governance

The completed expert tables, associated assumptions and resulting cut-off analyses must be archived.

The archive should include:

- expert tables;
- business rationales;
- benchmark studies;
- scenario analyses;
- cut-off simulations;
- meeting minutes;

- validation reports.

These artefacts should be available for:

- internal review;
- independent validation;
- periodic monitoring;
- supervisory requests.

4.2 Reject inference through score-based modelling

The objective of this section is to estimate the good/bad criterion for all applications in the study dataset when the target variable is unavailable.

The methodology described below is intended to:

- infer the unknown target;
- augment the study dataset;
- support cut-off construction;
- facilitate portfolio simulations.

4.2.1 Algorithm description

A score-based reject inference method assumes that the relationship between the score and the target observed on financed applications also holds, at least approximately, for rejected applications.

The methodology proceeds as follows:

1. Estimate a simplified good/bad model on the accepted population.
2. Use the model to compute an estimated probability of bad for applications with missing target.
3. Augment the original dataset with these estimated probabilities.
4. Recompute portfolio metrics and cut-off indicators on the enlarged population.

The simplified model may be a logistic regression:

$$Pr(bad) = \frac{1}{1 + \exp(a \times score + b)}$$

where:

- *score* is the original score;
- *a* and *b* are estimated coefficients.

4.2.2 Implementation notes

The implementation should guarantee:

- reproducibility;
- traceability;
- versioning of datasets and scripts;
- explicit assumptions;
- documented outputs.

The final dataset should contain:

- observed targets when available;
- inferred probabilities;
- flags distinguishing observed and inferred information;
- score buckets;
- additional variables used in simulations.

4.2.3 Application perimeter

The score-based reject inference approach should only be applied when:

- the accepted population is sufficiently representative;
- the score exhibits stable predictive power;
- the proportion of rejected applications remains reasonable;

- historical observations are available over a sufficiently long period.

This methodology is not recommended when:

- the score has recently changed significantly;
- rejection policies have evolved substantially;
- the portfolio composition is unstable;
- business practices have undergone major transformations.

Whenever score-based reject inference is applied, the following must be documented:

- assumptions;
- implementation choices;
- validation tests;
- sensitivity analyses;
- limitations.

5 Cut-off

The cut-off defines the acceptance threshold associated with the score.

Its objective is to balance:

- portfolio risk;
- business objectives;
- operational constraints;
- expected profitability.

Several cut-offs may coexist depending on:

- products;
- channels;
- countries;
- customer segments;
- business strategies.

Construction of the cut-off

The cut-off is generally determined using:

- cumulative acceptance rates;
- cumulative bad rates;
- expected profitability;
- business constraints;
- stress scenarios.

For a threshold c , define:

$$Acc(c) = \frac{\#\{score \geq c\}}{\#\{applications\}}$$

and

$$Bad(c) = \frac{\#\{bad ; score \geq c\}}{\#\{score \geq c\}}$$

The selected cut-off should provide:

- an acceptable acceptance rate;
- a satisfactory risk profile;
- operational feasibility;
- consistency with strategic objectives.

Sensitivity analyses

Several scenarios should be analysed.

Examples include:

- optimistic scenario;
- central scenario;
- conservative scenario;
- stressed macroeconomic scenario.

The objective is to evaluate:

- cut-off stability;
- portfolio sensitivity;
- expected profitability;
- robustness of business decisions.

Monitoring

After deployment, the cut-off should be monitored periodically.

Suggested indicators:

- acceptance rate;
- average score;
- bad rate;
- override rate;
- portfolio composition;
- comparison with simulations.

Any significant deviation should trigger:

- further investigations;
- scenario reviews;
- possible recalibration.

6 Paragraphe

This chapter is intentionally generic and may be adapted depending on local requirements.

It can contain:

- business considerations;
- operational constraints;
- governance principles;
- score deployment recommendations;
- specific portfolio analyses.

6.1 Sous-Paragraphe

Additional information may include:

- examples;
- use cases;
- practical recommendations;
- implementation notes.

The objective is to ensure:

- consistency of implementation;
- knowledge sharing;
- reproducibility;
- auditability.

7 Annexe

The annexes may contain:

- definitions;
- examples;
- benchmark studies;
- supplementary tables;
- technical notes;
- validation results.

7.1 Sous-Paragraphe

Example of a score development workflow:

Raw data ↓ Data preparation ↓ Variable selection ↓ Target definition ↓ Score estimation ↓ Validation

The complete study package should include:

- datasets;
- scripts;
- reports;
- scorecard documentation;
- validation outputs;
- monitoring material;
- governance documents.

End of document

Methodology for statistical scorecards
Version 7.2